

This is CS50

CS50's Introduction to Computer Science

OpenCourseWare

Donate [🔗](https://cs50.harvard.edu/donate) (<https://cs50.harvard.edu/donate>)

David J. Malan (<https://cs.harvard.edu/malan/>)

malan@harvard.edu

[f](https://www.facebook.com/dmalan/) (<https://www.facebook.com/dmalan/>) [G](https://github.com/dmalan/) (<https://github.com/dmalan/>) [@](https://www.instagram.com/davidjmalan/) (<https://www.instagram.com/davidjmalan/>) [in](https://www.linkedin.com/in/malan/) (<https://www.linkedin.com/in/malan/>) [@](https://www.reddit.com/user/davidjmalan/) (<https://www.reddit.com/user/davidjmalan/>) [@](https://www.threads.net/@davidjmalan/) (<https://www.threads.net/@davidjmalan/>) [🐦](https://twitter.com/davidjmalan/) (<https://twitter.com/davidjmalan/>)

Mario



Problem to Solve

In a file called `mario.py` in a folder called `sentimental-mario-less`, write a program that recreates a half-pyramid using hashes (`#`) for blocks, exactly as you did in [Problem Set](#)

1. Your program this time should be written in Python!

Demo

Specification

- To make things more interesting, first prompt the user with `get_int` for the half-pyramid's height, a positive integer between `1` and `8`, inclusive.
- If the user fails to provide a positive integer no greater than `8`, you should re-prompt for the same again.
- Then, generate (with the help of `print` and one or more loops) the desired half-pyramid.
- Take care to align the bottom-left corner of your half-pyramid with the left-hand edge of your terminal window.

How to Test

While `check50` is available for this problem, you're encouraged to first test your code on your own for each of the following.

- Run your program as `python mario.py` and wait for a prompt for input. Type in `-1` and press enter. Your program should reject this input as invalid, as by re-prompting the user to type in another number.
- Run your program as `python mario.py` and wait for a prompt for input. Type in `0` and press enter. Your program should reject this input as invalid, as by re-prompting the user to type in another number.

- Run your program as `python mario.py` and wait for a prompt for input. Type in `1` and press enter. Your program should generate the below output. Be sure that the pyramid is aligned to the bottom-left corner of your terminal, and that there are no extra spaces at the end of each line.

```
#
```

- Run your program as `python mario.py` and wait for a prompt for input. Type in `2` and press enter. Your program should generate the below output. Be sure that the pyramid is aligned to the bottom-left corner of your terminal, and that there are no extra spaces at the end of each line.

```
#
##
```

- Run your program as `python mario.py` and wait for a prompt for input. Type in `8` and press enter. Your program should generate the below output. Be sure that the pyramid is aligned to the bottom-left corner of your terminal, and that there are no extra spaces at the end of each line.

```
  #
  ##
 ###
####
#####
#####
#####
#####
#####
```

- Run your program as `python mario.py` and wait for a prompt for input. Type in `9` and press enter. Your program should reject this input as invalid, as by re-prompting the user to type in another number. Then, type in `2` and press enter. Your program should generate the below output. Be sure that the pyramid is aligned to the bottom-left corner of your terminal, and that there are no extra spaces at the end of each line.

```
#
##
```

- Run your program as `python mario.py` and wait for a prompt for input. Type in `foo` and press enter. Your program should reject this input as invalid, as by re-prompting the user to type in another number.
- Run your program as `python mario.py` and wait for a prompt for input. Do not type anything, and press enter. Your program should reject this input as invalid, as by re-prompting the user to type in another number.

Correctness

```
check50 cs50/problems/2025/x/sentimental/mario/less
```

Style

```
style50 mario.py
```

How to Submit

```
submit50 cs50/problems/2025/x/sentimental/mario/less
```

